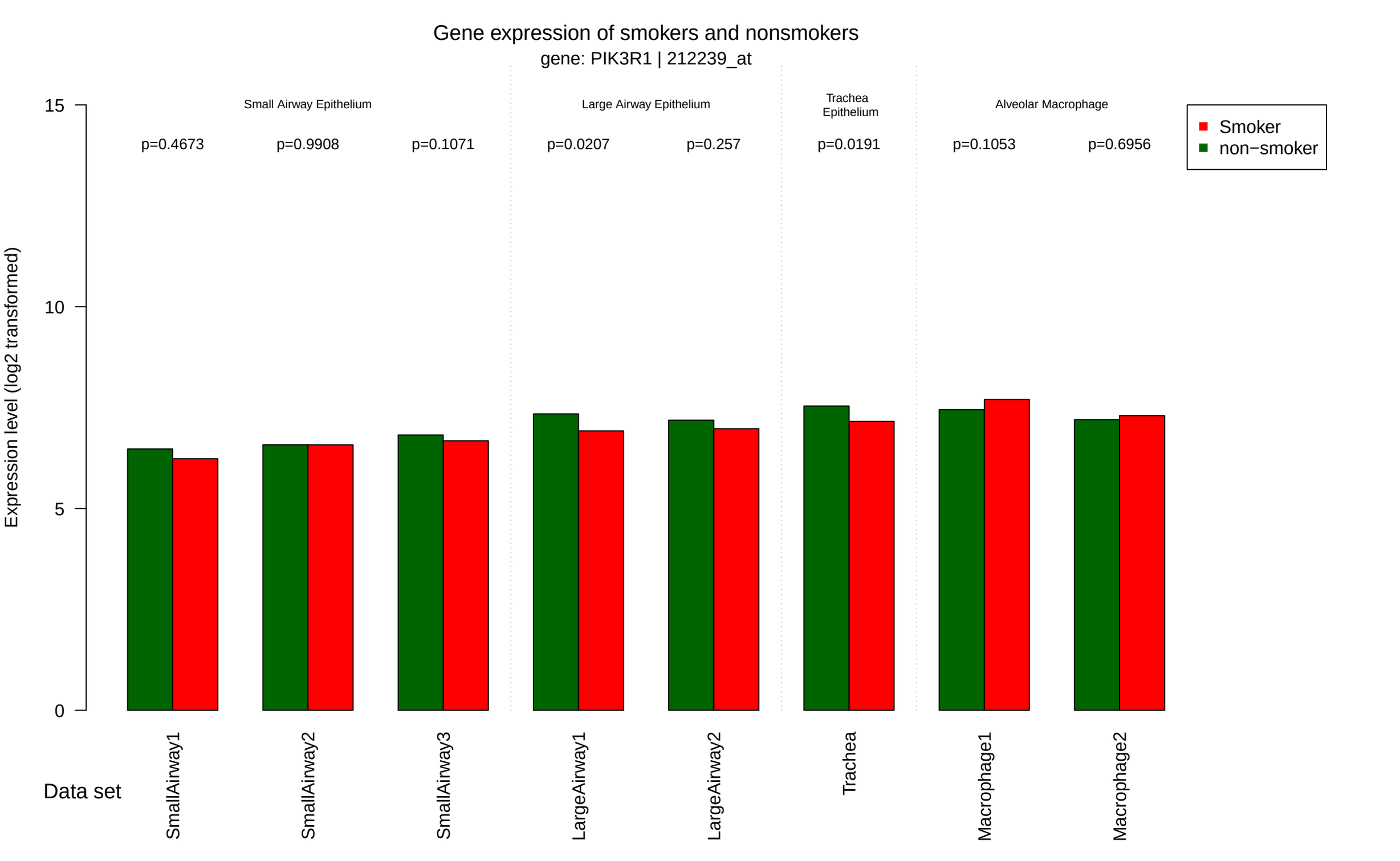
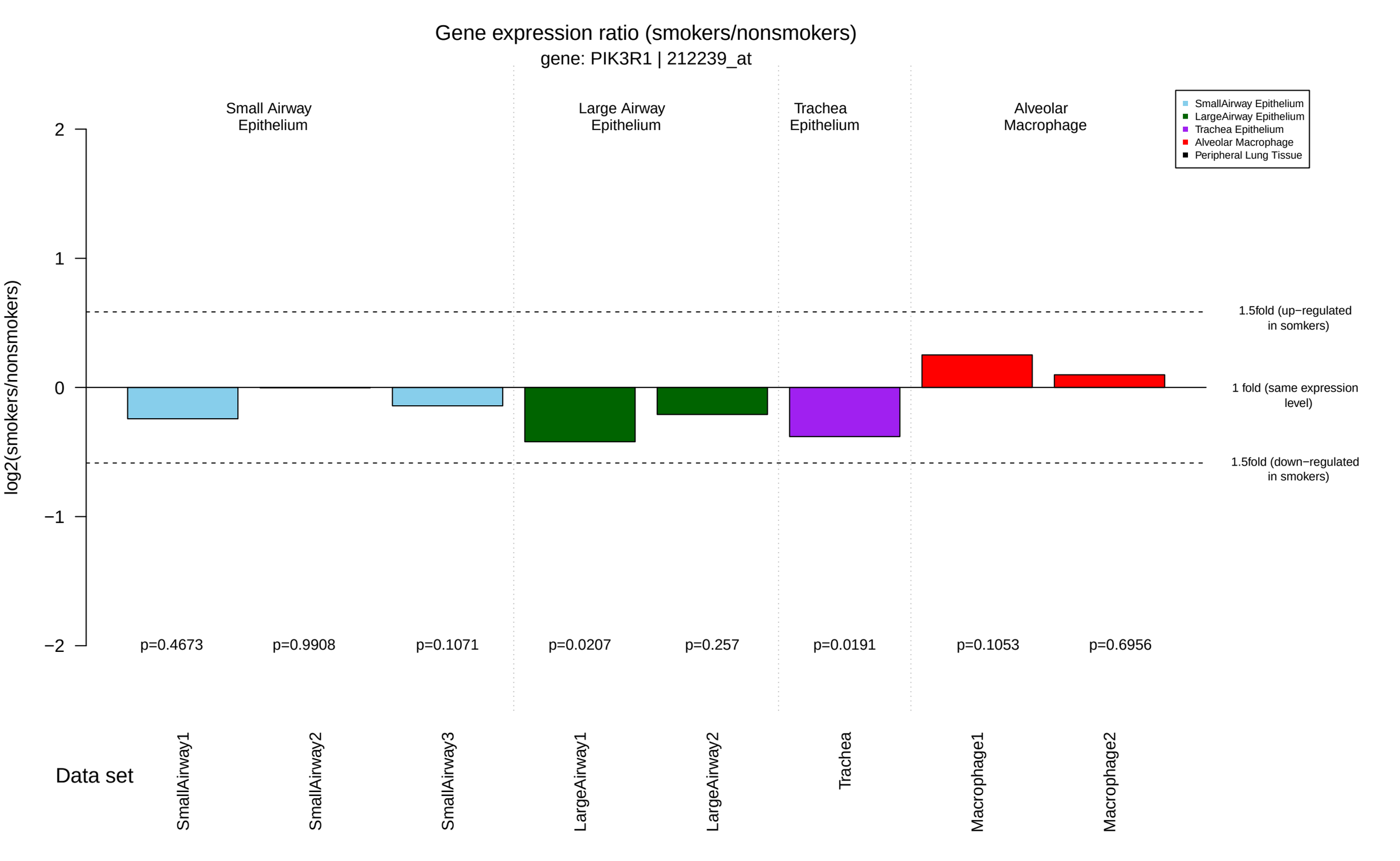
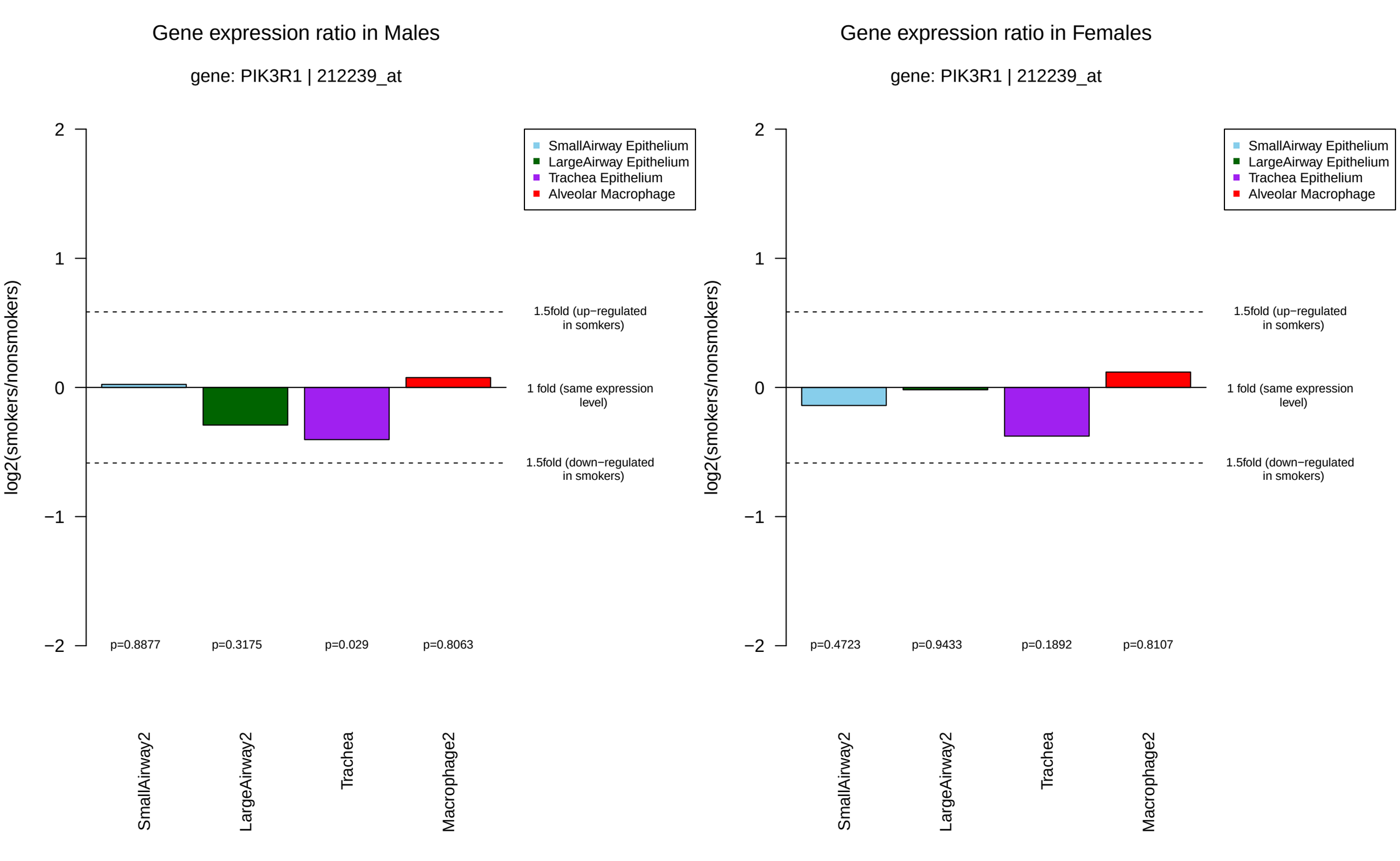
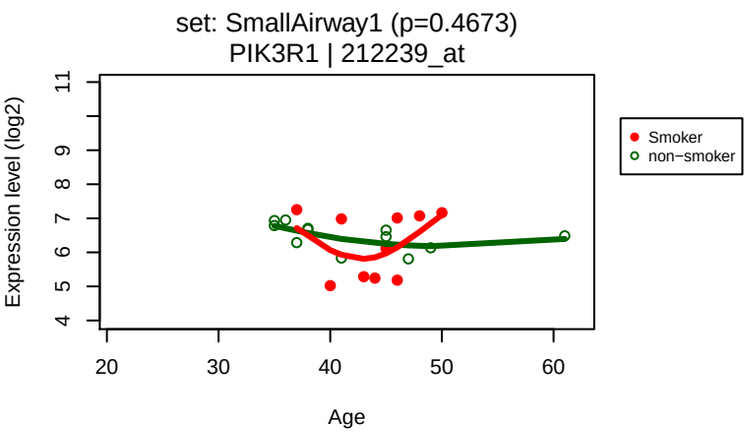
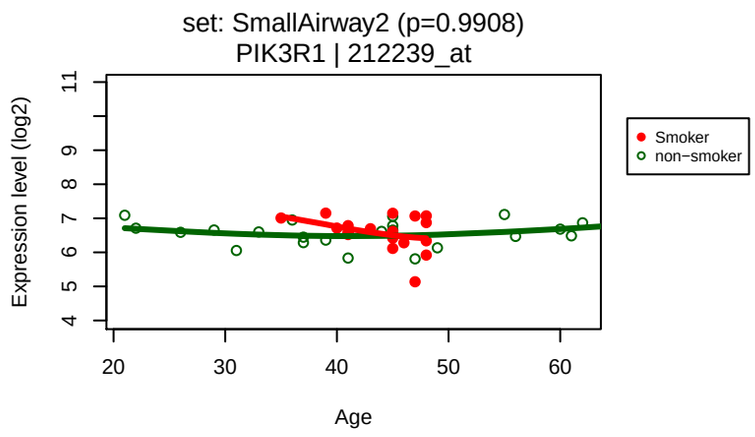
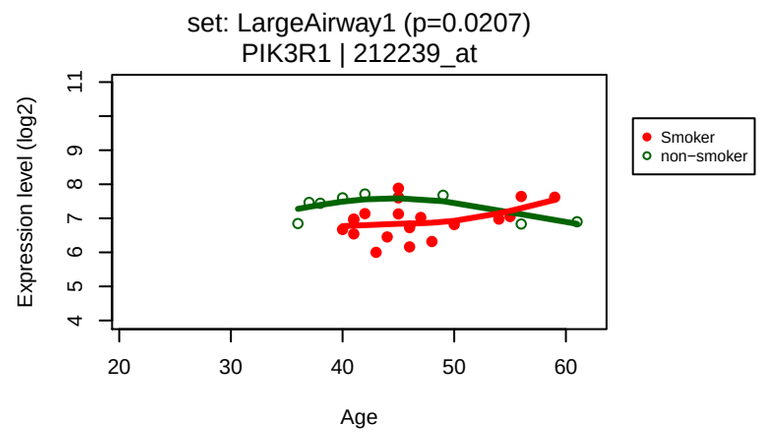

Gene expression of smokers and nonsmokers
gene: PIK3R1 | 212239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4673
p=0.9908
p=0.1071
p=0.0207
p=0.257
p=0.0191
p=0.1053
p=0.6956
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

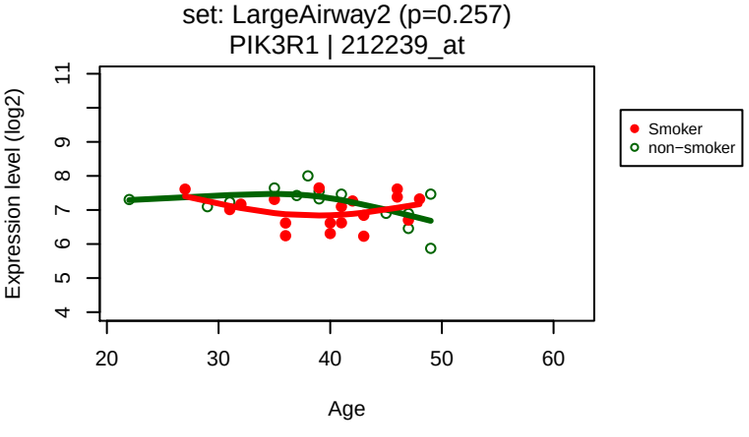

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R1 | 212239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4673
p=0.9908
p=0.1071
p=0.0207
p=0.257
p=0.0191
p=0.1053
p=0.6956
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

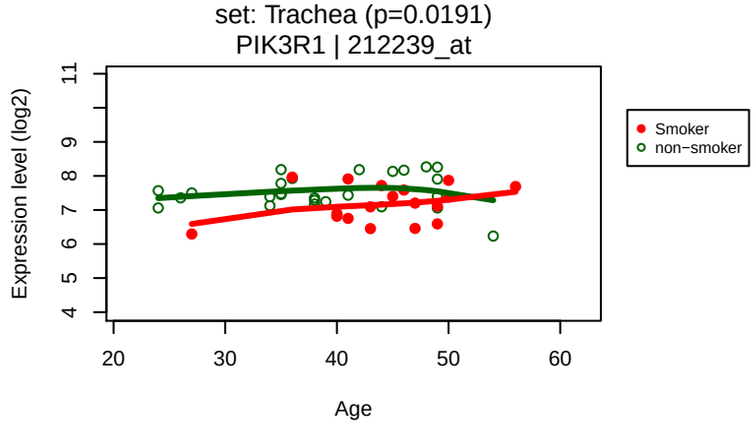

Gene expression ratio in Males
gene: PIK3R1 | 212239_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8877
p=0.3175
p=0.029
p=0.8063
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R1 | 212239_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4723
p=0.9433
p=0.1892
p=0.8107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

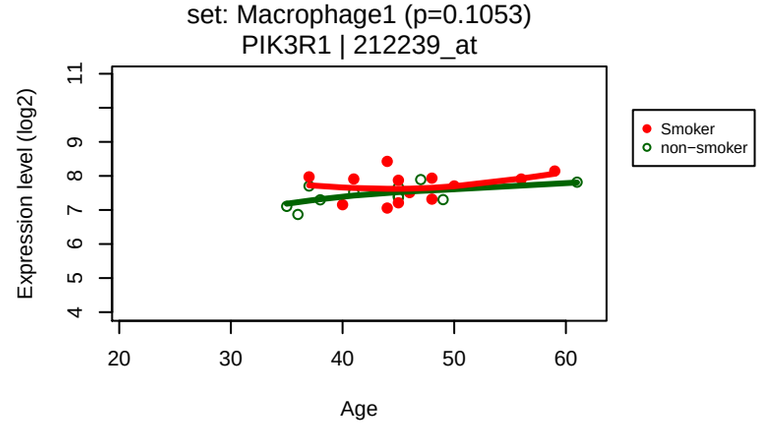

set: SmallAirway1 (p=0.4673)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker

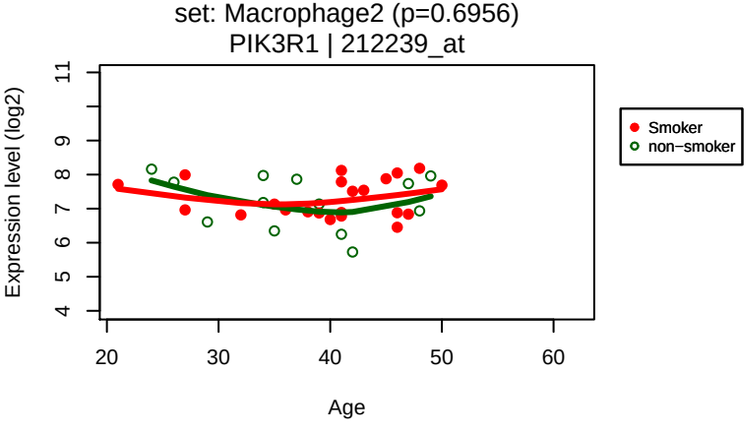

set: SmallAirway2 (p=0.9908)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker

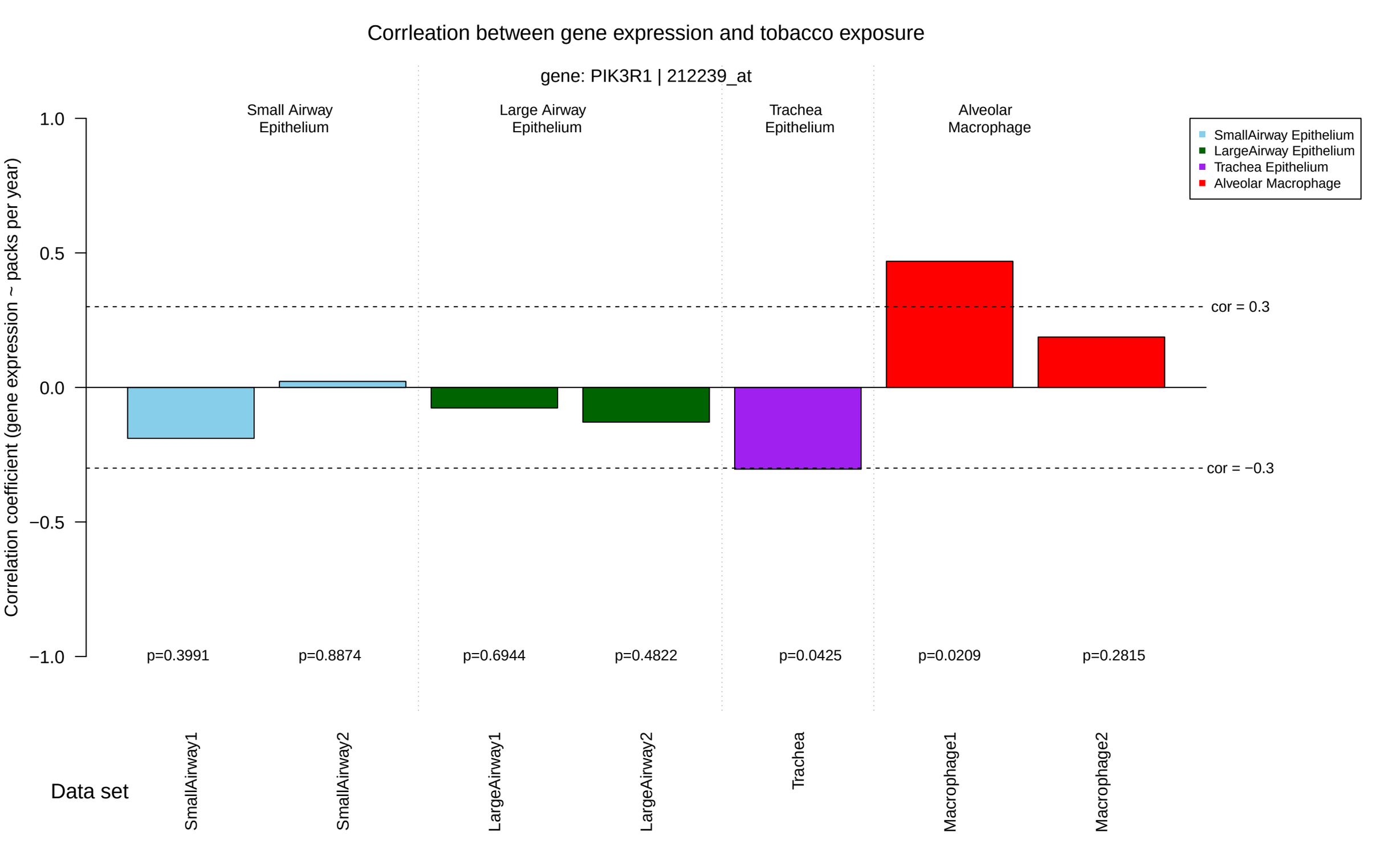

set: LargeAirway1 (p=0.0207)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.257)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0191)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1053)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6956)
PIK3R1 | 212239_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R1 | 212239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3991
p=0.8874
p=0.6944
p=0.4822
p=0.0425
p=0.0209
p=0.2815
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2